VIETNAM NATIONAL UNIVERSITY HCMC

INTERNATIONAL UNIVERSITY

School of Business

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REPORT

GROUP ASSIGNMENT

STATISTICS FOR BUSINESS

BUSINESS PERFORMANCE ANALYSIS & STATISTICAL RECOMMENDATIONS

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Group 6

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I. INTRODUCTION & DATA PREPARATION

1. Research objectives and context

In an increasingly competitive environment, businesses can no longer rely solely on intuition when making decisions; instead, they must leverage data analysis and statistical tools. Understanding the factors that affect revenue and especially profit is critical to improving operational performance and reducing risk.

The objectives of this study are to:

- Analyze historical sales data of a multi-channel retail business.
- Identify the key factors influencing profit, including pricing, discounting, and operational elements.
- Provide managerial recommendations based on statistical evidence (data-driven decision making).

The sample business in this study operates in the retail sector and sells three main product categories:

- Furniture
- Office Supplies
- Technology

Throughout this report, ***profit*** is selected as the primary performance indicator because high sales revenue does not necessarily mean business effectiveness if costs and discounts are not properly controlled.

2. Dataset overview

The original data were provided as a raw sales transaction file covering multiple years. Each row represents a single order, with the following main variable groups:

Quantitative variables:

- ***Sales:*** Revenue of the order
- ***Profit:*** Profit of the order

Discount: Discount rate applied

Shipping Cost: Shipping cost

Categorical variables:

Region: Sales region

Ship Mode: Shipping method (Regular Air, Express Air, Delivery Truck)

Product Category: Category (Furniture, Office Supplies, Technology)

The raw dataset reflects the company's large operating scale; however, it is not immediately ready for statistical analysis due to multiple data quality issues.

Role of the data in analysis

This dataset enables us to:

Analyze sales and profit over time, by region, and by product category.

Evaluate the relationships between discounting, shipping cost, and profit performance.

Conduct statistical tests to support business decision-making.

3. Data quality assessment & Outlier detection (Using Z-core and IQR)

Before performing statistical analysis, the data were examined and processed to ensure the reliability and validity of subsequent results.

3.1 Using IQR

Column: 'Order Quantity'

Q1 (25th percentile): 13.00

Q3 (75th percentile): 38.00

IQR: 25.00

Lower Bound: -24.50

Upper Bound: 75.50

Number of IQR Outliers: 0

Column: 'Sales'

Q1 (25th percentile): 143.50

Q3 (75th percentile): 1704.58

IQR: 1561.08

Lower Bound: -2198.12

Upper Bound: 4046.20

Number of IQR Outliers: 1039

Column: 'Discount'

Q1 (25th percentile): 0.02

Q3 (75th percentile): 0.08

IQR: 0.06

Lower Bound: -0.07

Upper Bound: 0.17 Number
of IQR Outliers: 3

Column: 'Profit'

Q1 (25th percentile): -83.32

Q3 (75th percentile): 162.79

IQR: 246.11

Lower Bound: -452.49

Upper Bound: 531.96

Number of IQR Outliers: 1704

Column: 'Unit Price'

Q1 (25th percentile): 6.48

Q3 (75th percentile): 85.99

IQR: 79.51

Lower Bound: -112.78

Upper Bound: 205.25

Number of IQR Outliers: 850

3.2 Using Z-core

Column: 'Order Quantity'

Number of Z-score Outliers (|Z| > 3): 0

No Z-score outliers found for this column with |Z| > 3.

Column: 'Sales'

Number of Z-score Outliers (|Z| > 3): 206

Column: 'Profit'

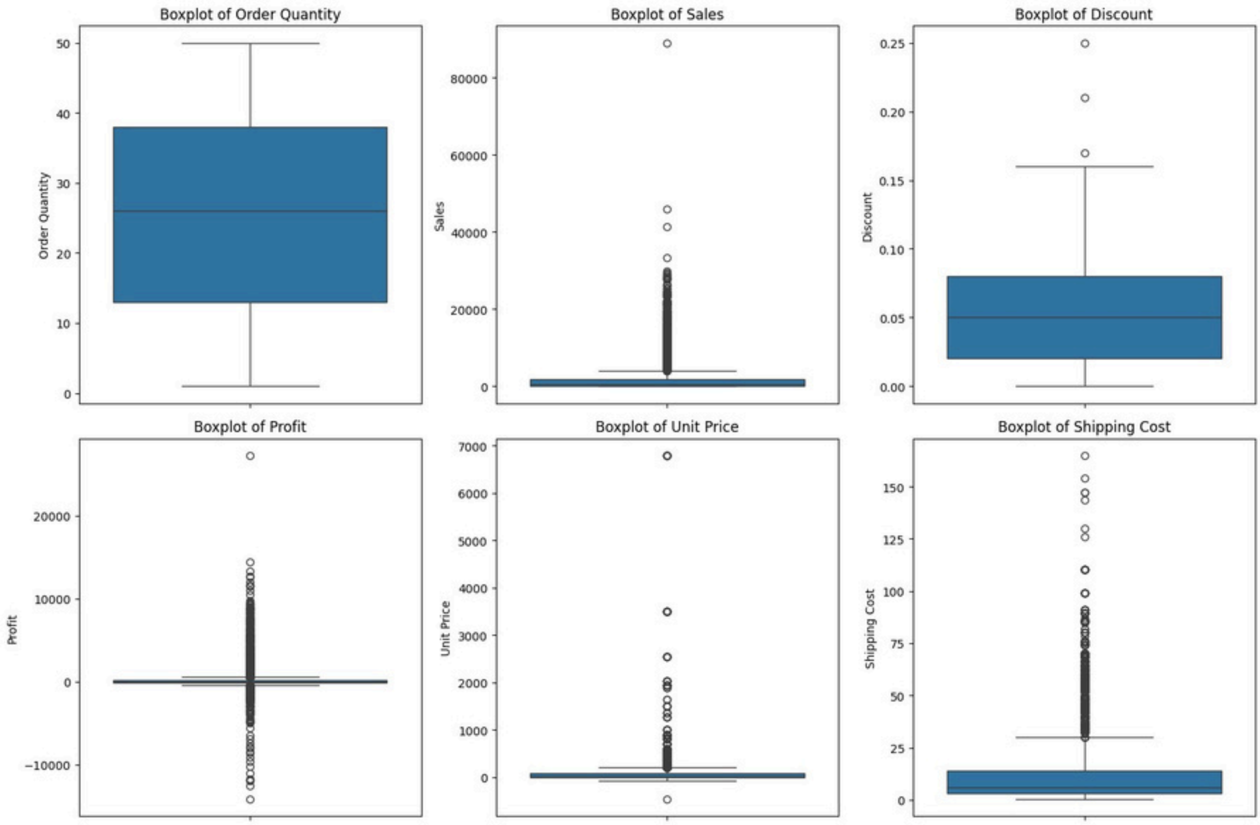
Number of Z-score Outliers (|Z| > 3): 178

Column: 'Unit Price'

Number of Z-score Outliers (|Z| > 3): 64

□ ***Column: 'Shipping Cost'***

Number of Z-score Outliers (|Z| > 3): 229



□ ***Visualizing Outliers with Boxplots***

3.3 Analytics

**** Order Quantity***

The distribution of order quantity is relatively balanced, with the median close to the average level. There are no extreme outliers observed.

□ This indicates that order volumes are fairly stable, with few unusually small or large orders.

**** Sales***

Sales exhibit a ***strong right-skewed distribution*** with a large number of high-value outliers. A small number of orders generate exceptionally high revenue compared to the majority.

□ This suggests that total sales are heavily driven by a limited number of high-value transactions, which should be analyzed separately due to their significant impact on overall performance.

**** Discount***

Most discount values are concentrated at ***low levels (0–10%)***, while a few outliers appear at ***higher discount rates (above 20%)***.

□ This implies that deep discounting is not commonly applied and may occur only during special promotional campaigns.

**** Profit***

Profit shows a ***very wide distribution***, including both large positive profits and substantial losses. Numerous negative outliers indicate the presence of severely unprofitable orders.

□ This highlights the need to closely examine the relationship between ***discounts, costs, and profitability***, as high sales do not necessarily translate into high profit.

**** Unit Price***

Most products have relatively low unit prices, while a small number of items are priced extremely high, resulting in a right-skewed distribution with many outliers.

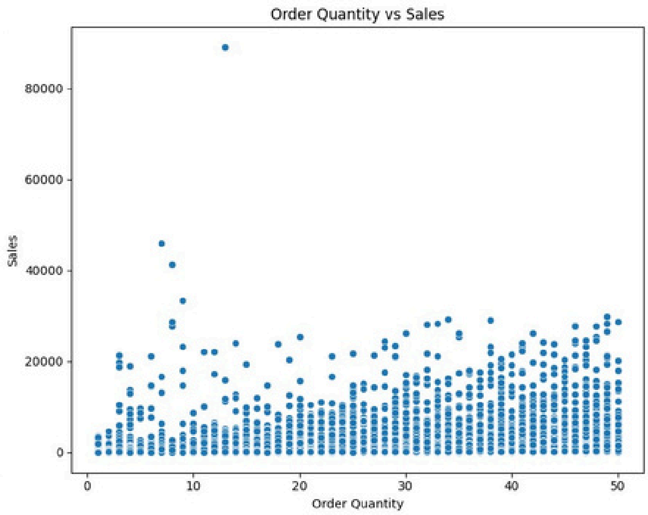
□ This reflects a diverse product portfolio, suggesting that analysis should be conducted by product segments rather than in aggregate.

**** Shipping Cost***

Shipping costs are generally concentrated at low to moderate levels, with several high-cost outliers. These may be caused by long-distance deliveries or bulky items.

□ Shipping costs represent a critical factor that can significantly erode profit margins if not properly managed.

Visualizing Bivariate Outliers with Scatter Plot



**** Sales vs Profit***

The scatter plot shows a ***generally positive relationship*** between sales and profit, indicating that higher sales tend to be associated with higher profits. However, the relationship is ***not linear***, and there is substantial dispersion in profit values at similar sales levels.

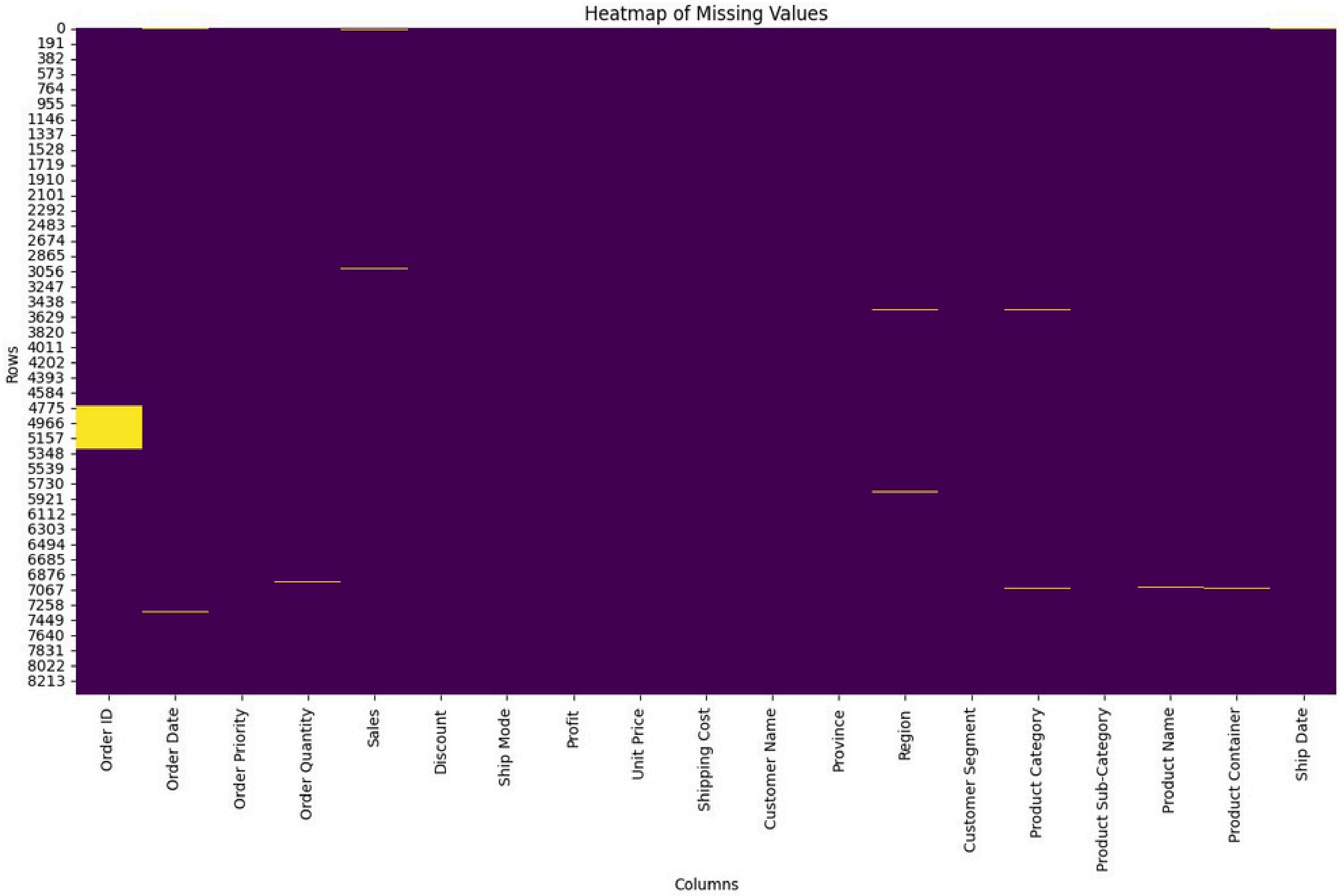
Notably, several data points display ***negative profit despite relatively high sales***, suggesting that high revenue does not guarantee profitability. This may be driven by factors such as excessive discounts, high shipping costs, or elevated production expenses. Additionally, a small number of extreme outliers generate exceptionally high profits and have a disproportionate impact on overall performance.

□ This highlights the importance of analyzing ***cost structures and discount strategies***, rather than relying solely on sales growth.

3.4 Missing values summary and visualization

	Missing Count	***Missing Percentage***
Order ID	545	6.488868
Sales	34	0.40481
Product Category	31	0.369092
Region	30	0.357185
Product Container	15	0.178593
Ship Date	14	0.166687
Product Name	13	0.15478
Order Date	12	0.142874
Order Priority	12	0.142874
Order Quantity	7	0.083343
Ship Mode	4	0.047625
Customer Name	1	0.011906
Profit	1	0.011906
Shipping Cost	1	0.011906

Visualizing Missing Data with Heatma:



 The heatmap provided a visual confirmation of the distribution of missing values, indicating that missingness is scattered across various columns and rows rather than being concentrated in specific blocks, suggesting that missing data may be random

3.5 Processing methods

Issues Detected in the Raw Data

Some rows have inconsistent date formats or year values that do not align with the research period.

Some quantitative variables (Sales, Profit, Discount, Shipping Cost) have incorrect data types (stored as text).

Some observations contain missing or invalid values.

The data cleaning steps were conducted as follows:

Remove invalid observations and rows with severe errors.

Convert quantitative variables to numeric format for statistical computation.

During cleaning, we found that the year 2022 contained only 16 transactions recorded on a single day (11 December 2022**).** This is incomplete data and could distort time-trend analysis. Therefore, we removed all 2022 records and focused on the stable period 2009–2012**.**

Recheck the dataset after cleaning to ensure there are no missing values or formatting issues that could affect the analysis.

After cleaning, the final dataset used for analysis:

Accurately reflects the study period 2009–2012**.**

It is suitable for inferential statistical analyses in the following sections.

II. SALES PERFORMANCE ANALYSIS

1. Sales Overview

1.1 Data Scope and Revenue Scale

To ensure accuracy and consistency for trend analysis, the data cleaning process excluded incomplete observations from 2022. The final dataset represents a continuous 4-year business cycle (2009–2012) with the following core metrics:

Valid Transactions (n): 8,210 orders.

Total Revenue: $14,554,287.55.

Assessment: This revenue figure reflects significant operational scale in the multi-category retail sector. Standardizing the data eliminates noise (such as the fragmented 2022 data), ensuring high reliability when evaluating growth models and seasonality.

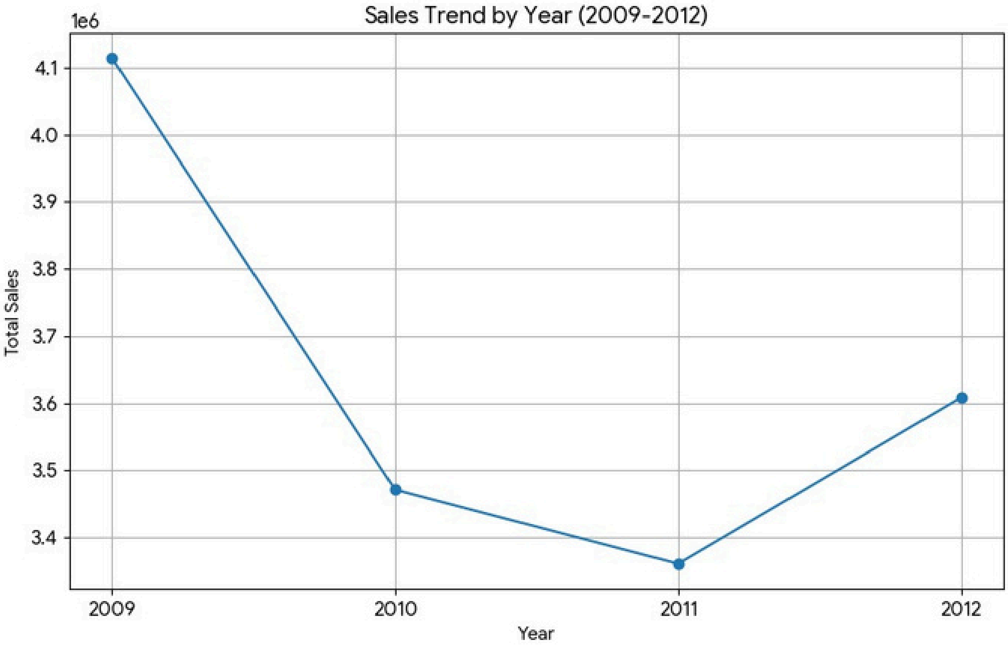
1.2 Year-Over-Year Trend Analysis

Revenue data across fiscal years reveals a notable fluctuation pattern:

Order Year	***Total Sales ($)***	***Share***	***Trend***
2009	4,113,999.11	28.27%	***Growth Peak***
2010	3,471,205.76	23.85%	Sharp Decline (-15.6%)
2011	3,360,899.64	23.09%	Slight Decline

2012	3,608,183.04	24.79%	Recovery (+7.4%)

Deep Dive Analysis: The business experienced a "U-shaped" volatility cycle.



Decline Phase (2010–2011): Following a peak in 2009 (reaching over $4.1 million), the company experienced a severe downturn over the subsequent two years. Revenue dropping to the $3.3 – $3.4 million range suggests the potential loss of key accounts or negative impacts from the macroeconomic environment.

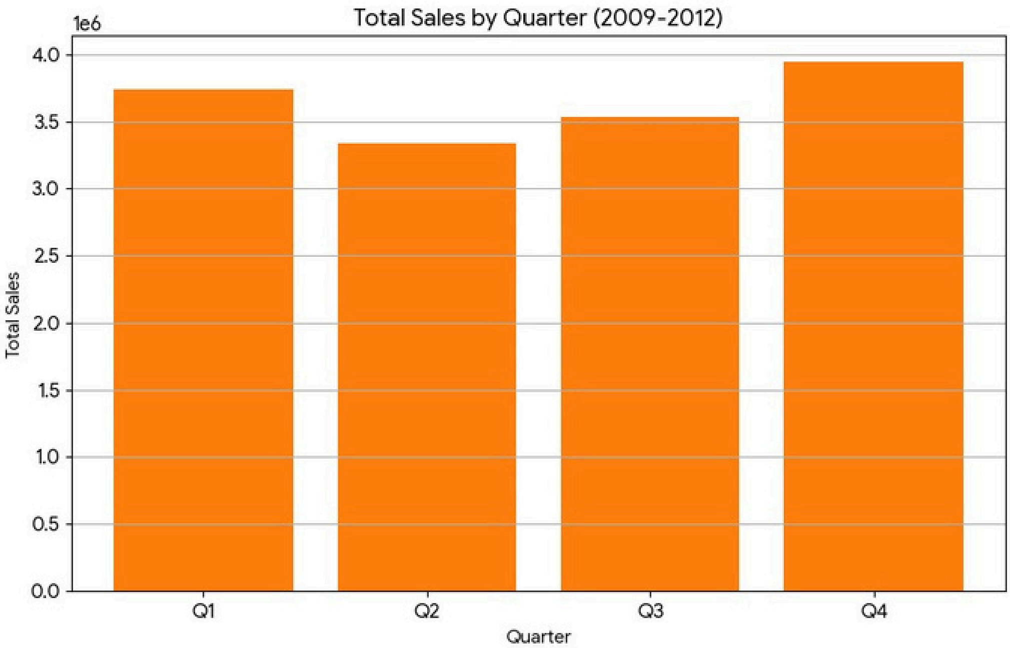
Recovery Phase (2012): 2012 marked a return to growth. However, the revenue of $3.6 million remains below the 2009 peak.

Conclusion: The revenue state in the last 3 years (2010–2012) can be described as ***"stagnation."*** The enterprise is struggling to find new growth drivers and has not yet broken out of its safety zone. This necessitates stronger restructuring solutions at the regional and category levels.

1.3 Seasonality Analysis

Quarterly revenue verification over the 4 years shows clear purchasing behavior patterns

Quarter	*Total Sales ($)*	*Characteristics*
Q1	3,739,959.41	Stable
Q2	3,333,977.64	Trough (Lowest)
Q3	3,534,587.36	Gathering momentum
Q4	3,945,763.15	Peak Season



Deep Dive Analysis: The data confirms strong ***Seasonality***:

Year-End Effect: Q4 is consistently the sales boom period, significantly outperforming Q2 (the low season). This is typical in retail as demand surges during holidays and corporate year-end budget clearing cycles.

Managerial Implication: Q2 revenue being lower than Q4 is a normal cyclical phenomenon. However, it mandates cash flow management: The company needs to accumulate resources in Q2-Q3 to meet the order explosion in Q4.

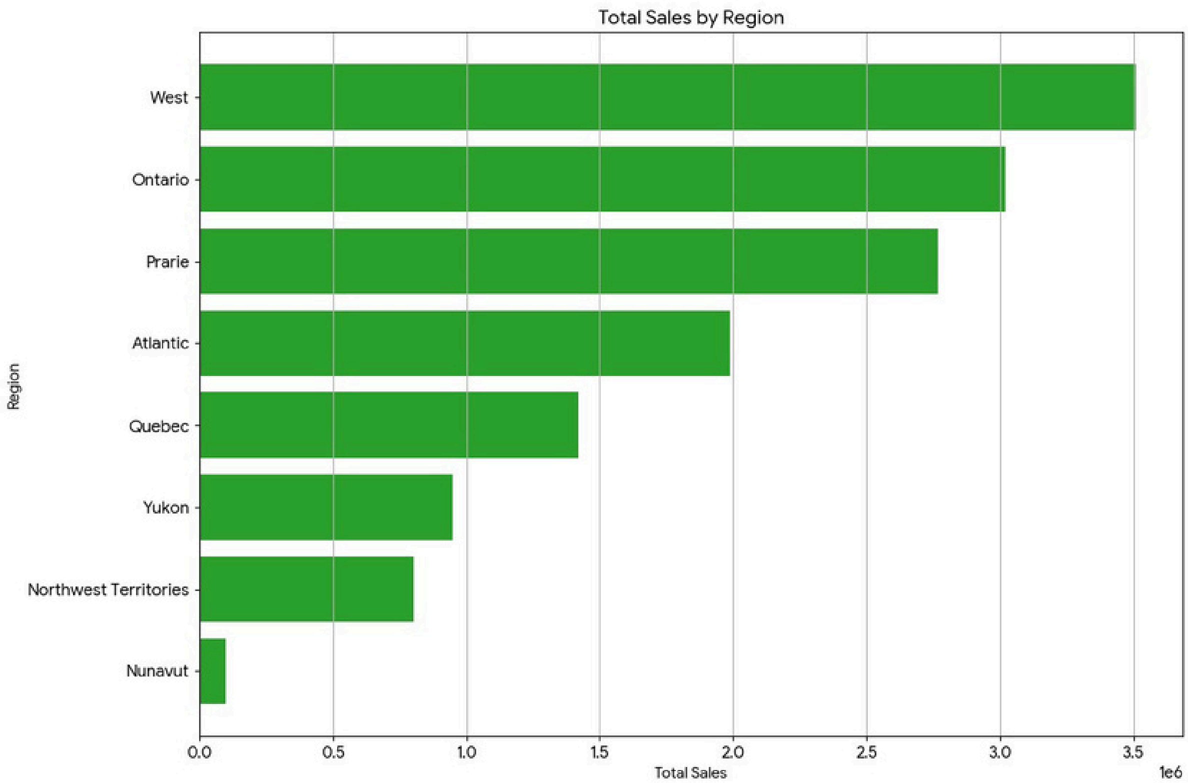
Conclusion: There is a strong dependency on Q4 (Year-end effect), while Q2 consistently underperforms. This creates uneven cash flow and operational pressure at the end of the year.

2. Sales Decomposition & Market Structure

2.1 Regional Performance Analysis

The disparity in revenue contribution across geographic regions indicates a high degree of centralization:

Strategic Tier	***Region***	***Total Sales ($)***	***Share***	***Assessment***
Core Markets	***West***	3,510,012	24.12%	***Critical***
	Ontario	3,020,460	20.75%	***Critical***
	Prarie	2,766,581	19.01%	***Critical***
Satellite Markets	Atlantic	1,987,763	13.66%	Potential
	Quebec	1,420,087	9.76%	Potential
Niche Markets	Yukon, Northwest, Nunavut	~1.8M	~12%	Low Efficiency



Deep Dive Analysis:

The 80/20 Rule: The top three regions (***West, Ontario, Prarie***) contribute approximately ***64%*** of the company's total revenue. These form the financial "backbone" of the enterprise. Any negative fluctuation in these three regions will directly impact cash flow viability.

Low Efficiency Regions: Areas like Nunavut contribute less than 1% of revenue (yet likely incur very high shipping costs due to geography).

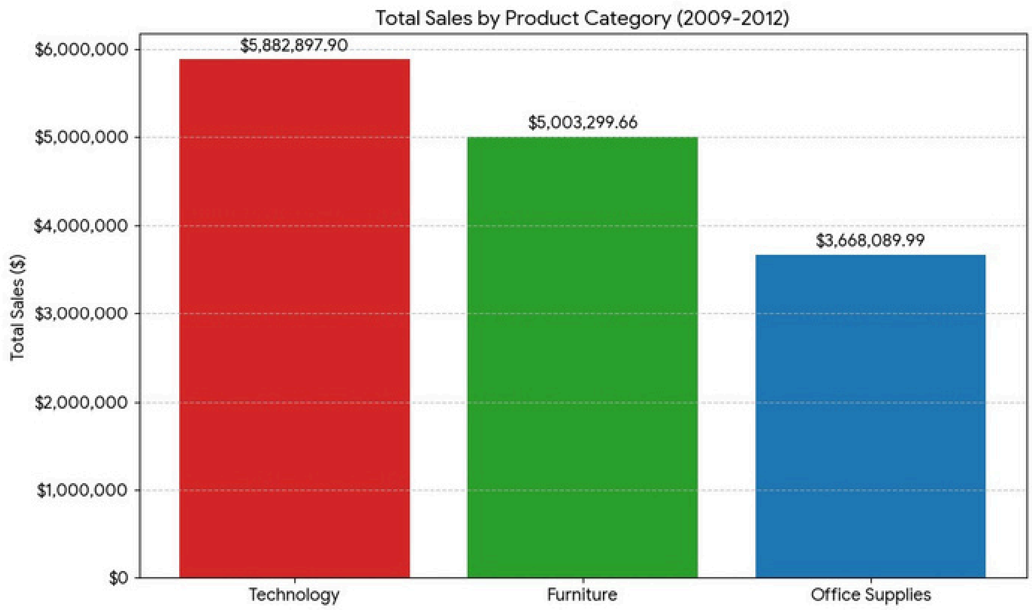
Strategic Recommendation: The company should adopt a ***"Resource Concentration"*** strategy. Prioritize marketing budgets and incentive policies for the Core Markets group to maximize ROI, rather than spreading resources thin across low-efficiency niche markets like Nunavut.

Conclusion: The company follows the 80/20 Rule, where three key regions drive the majority of revenue. Remote areas like Nunavut contribute less than 1% but likely incur disproportionately high shipping costs.

2.2 Category Portfolio Analysis

Revenue structure by main product group reveals potential profit risks:

Category	***Total Sales ($)***	***Share***
Technology	5,882,897.90	***40.42%***
Furniture	5,003,299.66	34.38%
Office Supplies	3,668,089.99	25.20%



Deep Dive Analysis:

Dominance of High-Ticket Items: Technology and Furniture account for nearly 75% of total revenue. This indicates the company's business model relies on selling expensive products rather than high volumes of small items.

"Revenue Trap" Warning: While these categories bring in large cash flows, they often carry high Cost of Goods Sold (COGS) and operational costs (Shipping, Warehousing). This lays the groundwork for examining true profitability efficiency in Part 3.

Conclusion: The model relies on High-Ticket items (Tech/Furniture). While this generates high revenue, it risks the "Revenue Trap" where high operational costs (shipping bulky furniture) eat into profits. Office Supplies has low revenue but high transaction frequency.

2.3 Top revenue products

To delve into product-level details, the research team identified the Top 5 products (SKUs) with the highest total sales during the analysis period:

Rank	***Product Name***	***Total Sales ($)***	***Category***
1	Polycom ViewStation™ ISDN Videoconferencing Unit	255,303.97	Technology
2	Global Troy™ Executive Leather Low-Back Tilter	252,701.56	Furniture
3	Canon PC940 Copier	210,910.68	Technology
4	Riverside Palais Royal Lawyers Bookcase	206,494.13	Furniture
5	Hewlett-Packard LaserJet 3310 Copier	194,880.35	Technology

Assessment & Analysis:

High-Ticket Item Dominance: The Top Products list is composed entirely of ***Technology*** (Conferencing units, Copiers) and ***Furniture*** (Executive chairs, Bookcases), aligning perfectly with the category revenue structure analyzed in section 2.2.

Risk Characteristics: These are all high-unit-value items. While high sales volume for these products generates impressive revenue streams, they often require significant discounts to stimulate B2B demand or incur high operational/warranty costs.

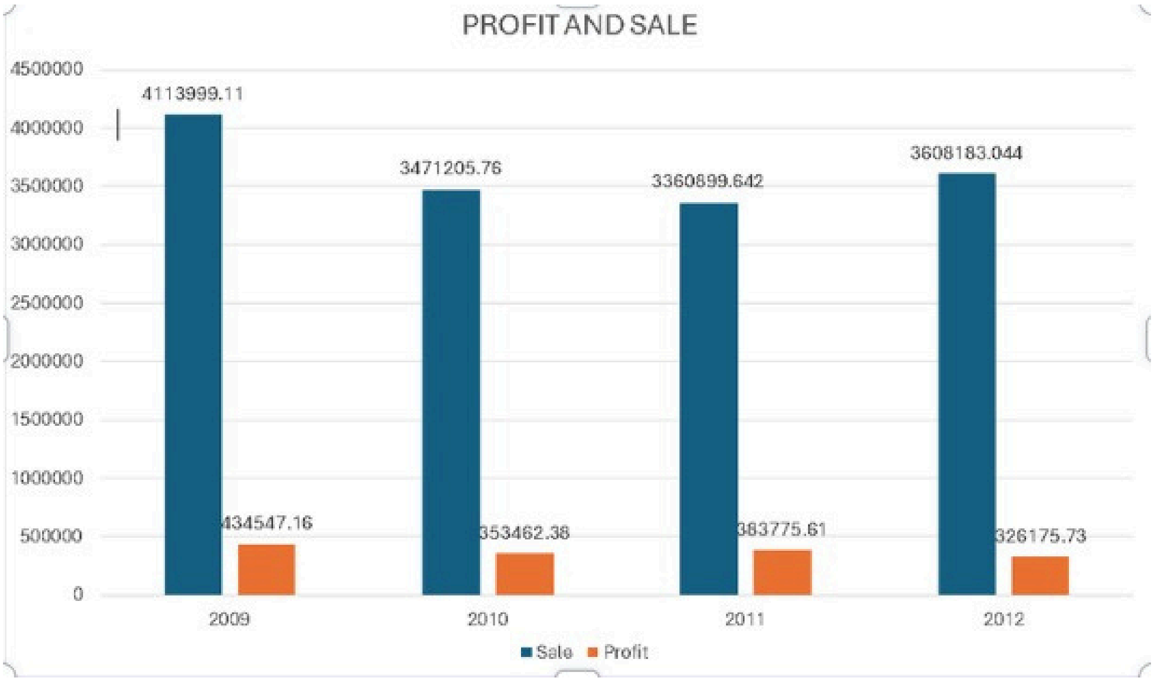
Conclusion: Revenue is heavily concentrated in a few high-value Technology and Furniture items. This creates ***Concentration Risk*** if these 5 items go out of stock, total revenue will plummet.

III. PROFIT ANALYSIS & RISKS

1. Profit Performance Overview

1.1 Financial Results Overview from Data

- ***Total Sales***: $14,554,287
- ***Total Profit***: $1,497,910
- ***Profit Margin***: 10.29%

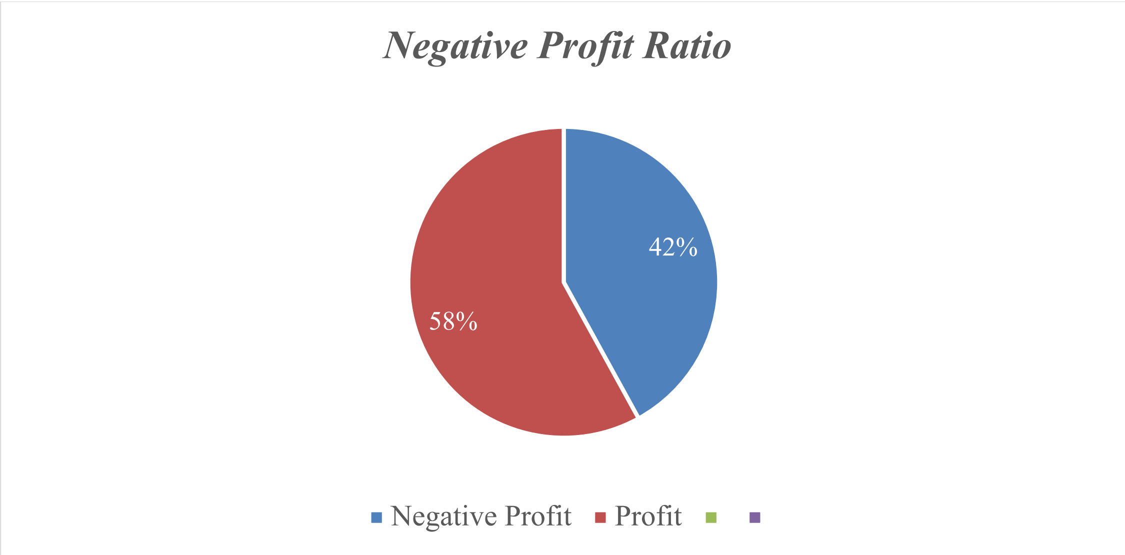


⇒Overall, the company is profitable with a margin of approximately 10.29%.

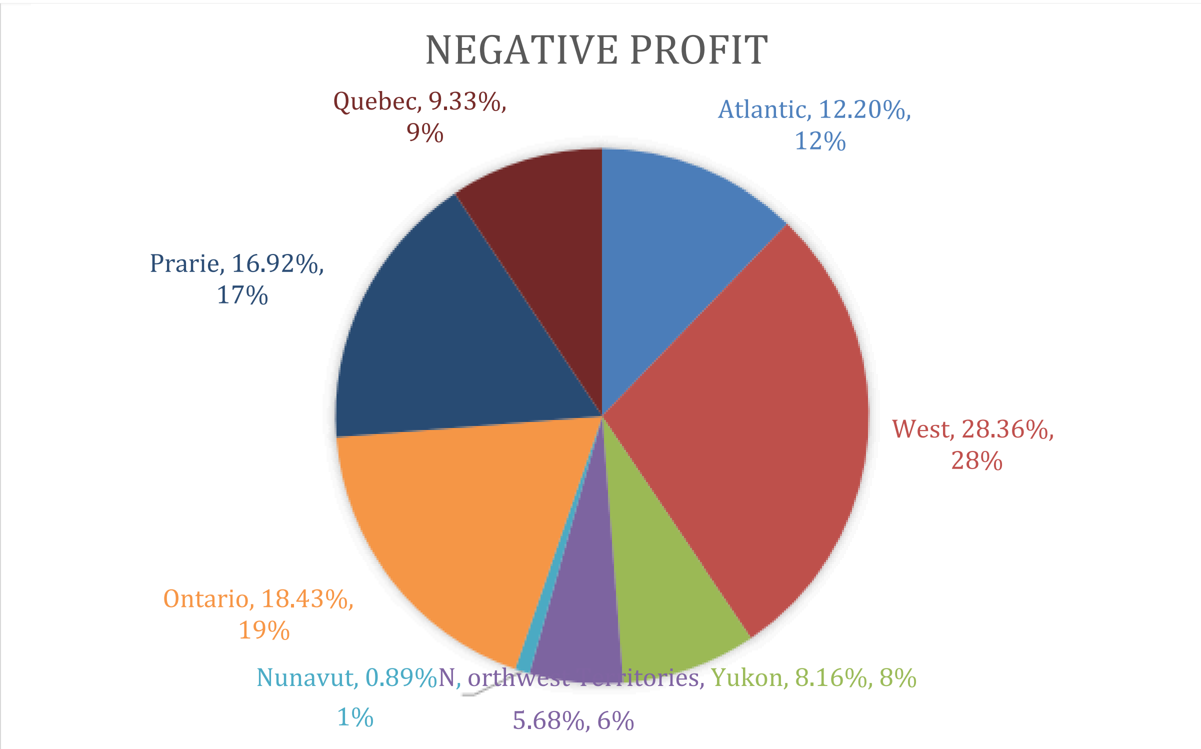
1.2 Major Risk: High Rate of Loss-Making Orders

o The percentage of transactions with Profit < 0 is about 50.78% (i.e., more than half of the transactions are loss-making).

Management Implication: Despite having positive overall profit, the profit structure is not sustainable (profits come from only a portion of orders, while many others are "burning" profits).



2. *Identify the Negative Profit*



- 70% of the total negative profit is concentrated in three major regions: West, Ontario, and Prairie.
- Some small regions (Nunavut, Northwest Territories) barely affect the total negative profit.
- The chart indicates that West is the most crucial region to focus on for profit improvement, and other large regions should also be reviewed.

3. Negative Profit Concentration

3.1 By Product Category:

Comparison of total Sales, total Profit, Profit Margin, and loss-making order percentage across 3 product categories:

Technology:

- o Sales ≈ $5,003,300
- o Profit ≈ $115,018
- o Margin ≈ 14.95%

Office Supplies:

- o Sales ≈ $3.67M
- o Profit ≈ $503,605
- o Margin ≈ 13.72%
- o Loss-Making Order Rate ≈ 53.44%

Furniture:

- o Sales ≈ $5,500,330
- o Profit ≈ $115,018
- o Margin ≈ 2.3% (very low)

Analysis of 3 Product Categories:

Technology – Most Profitable Category

- o Median = 67.36, the highest among the three categories, indicating that most Technology products generate positive profit.
- o IQR = 650.95, showing a wide profit variation across products.

o Despite the wide IQR, the number of outliers is relatively low (251), mainly positive, reflecting several high-margin “big win” products.

Furniture – Riskiest Category

o Median = –14.27, the lowest, meaning that 50% of Furniture products are unprofitable.

o IQR = 464.78, indicating high volatility.

o 293 outliers, mostly negative, suggesting substantial loss-making products.

Office Supplies – Stable but Low Profit

o Median = –3.94, slightly below zero, indicating small profits or break-even for many products.

o IQR = 114.56, the smallest, showing the most stable profit distribution.

Interpretation:

Furniture & Office Supplies: The negative median suggests that "typical orders" in these categories do not generate profit, though overall the company may still be profitable due to a few large orders.

Technology: The positive median and higher mean make it the "profit engine" for the company.

The wide variation in profit (with both very low minimums and high maximums) across all three categories indicates strong outliers. A boxplot should be used to clearly display this distribution.

Conclusion of Category Comparison:

Technology: The category with the best profit efficiency (high margin, positive median).

Furniture: The riskiest category (very low margin, negative median, and many large losses concentrated in Tables/Bookcases).

Office Supplies: Although it shows overall positive profit, its distribution is skewed, and the loss-making order percentage is high discount and related costs need to be better controlled.

3.1 By Product Sub - Category

Losses not just from "Furniture" generally, but heavily concentrated in ***Tables*** and ***Bookcases***.

- ***Tables (Furniture)***: Profit ≈ –$101,356.66 (very large loss)
- ***Bookcases (Furniture)***: Profit ≈ –$29,798.96

Conclusion: Losses are not just from "Furniture" generally, but heavily concentrated in Tables and Bookcases – these are the "hot spots" that need to be targeted.

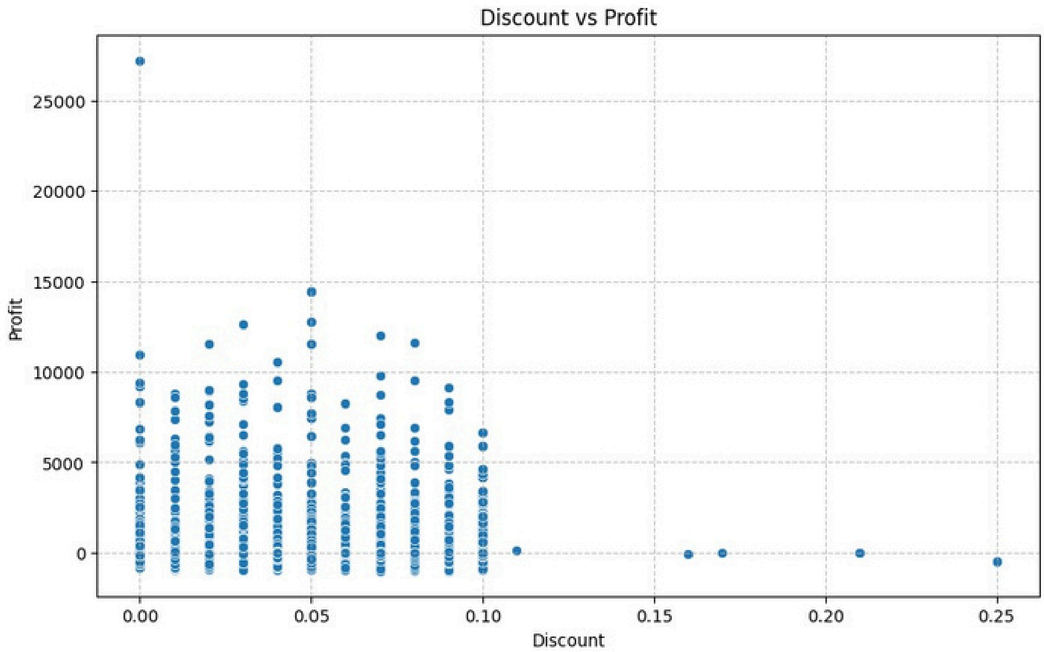
4. Discount vs. Profit: Correlation Analysis

4.1 Research Question

Does discounting (Discount) truly help the business, or does it erode profit?

To statistically examine the relationship between Discount and Profit:

Scatter plot (Discount – Profit)



- ***More discount, less profit***: As the discount rate (horizontal axis) increases (e.g., 15%, 20%, 25%), the green dots representing individual transactions tend to fall lower on the vertical axis (lower profit).
- ***Golden Point (0% - 5%)***: When no discount or a very small discount is applied, the company experiences extremely profitable orders (dots rise to the highest).

Danger Zone (Above 10%): When discounts exceed 10%, very few orders yield high profits, and some even become loss-making (dots below the 0 mark), especially when the discount reaches 25%.

4.2 Analysis by Discount Groups

0%: Total Margin ≈ 12.93%, Loss-Making Orders ≈ 45.99%
0–5%: Total Margin ≈ 12.08%, Loss-Making Orders ≈ 48.37%
5–10%: Total Margin ≈ 7.68%, Loss-Making Orders ≈ 54.23%
20%+: Total Margin ≈ –49.76% (significant loss) (small sample but a strong warning)

Quantitative Conclusion:

When Discount shifts from ≤5% → 5–10%, the total profit margin drops sharply (from around 12% to ~7.7%) and the percentage of loss-making orders increases.

5. Recommendations

Instead of relying on deep discounts that erode profit margins and brand value, the company should implement a discount threshold policy, where promotions are only applied when specific conditions are met.
Deep discounts should be replaced with alternative promotional strategies such as product bundling, conditional free shipping, and low-cost add-on gifts. These approaches help increase average order value, stimulate additional purchases, and enhance customer perceived value, while minimizing negative impacts on profitability.

IV. OPERATIONAL FACTORS: Analysis of shipping costs and their impact on profitability

1. Overview of Shipping Cost

From the cleaned data:

Total Shipping Cost: $105,548.62
Shipping Cost / Total Sales Ratio: ~0.7%

Conclusion: While shipping costs account for approximately ***0.7% of total sales***, the risk lies in the uneven distribution of shipping costs, which disproportionately affects low-value orders, leading to negative profit margins.

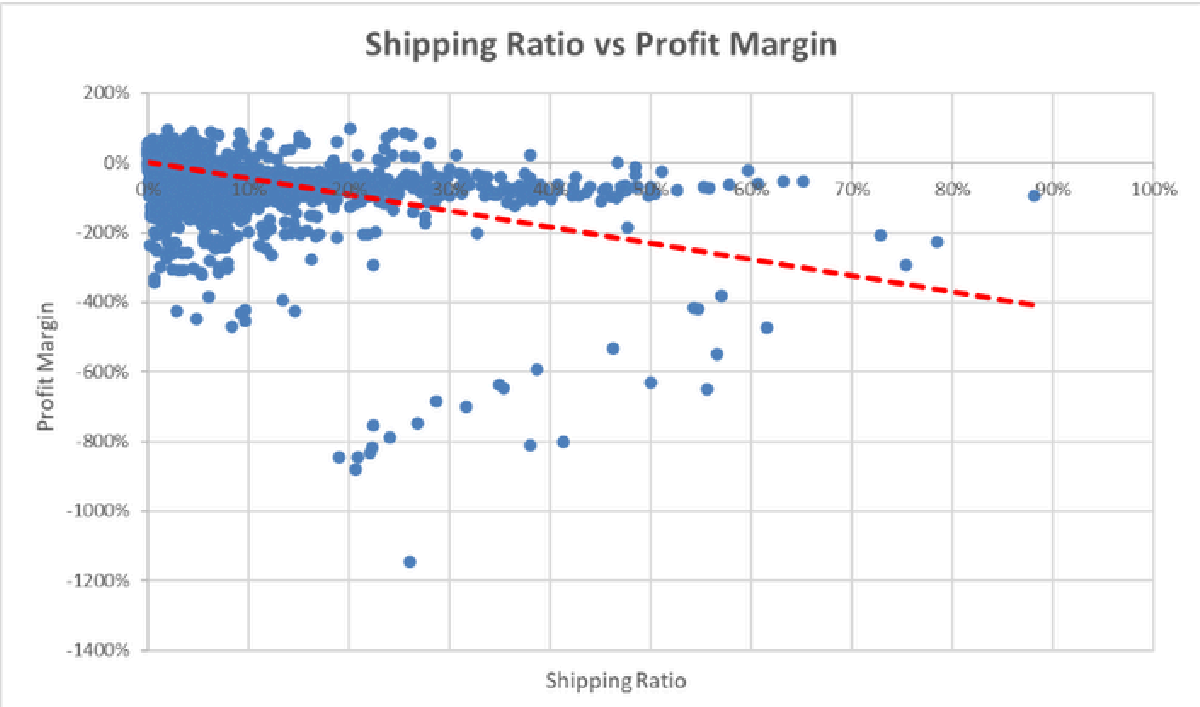
2. Key Operational Metric: Shipping-to-Sales Ratio

To measure the "weight" of shipping on each order, we use:

Relationship Analysis:

Correlation between ***Shipping Ratio*** and ***Profit Margin:*** r ≈ ***-0.481*** (strong negative correlation).

Statistical analysis shows that as the Shipping Ratio (shipping cost / sales) increases, profit margins tend to decrease. With a correlation coefficient of r ≈ ***-0.481***, there is a strong negative relationship between shipping costs and profit per order. However, this is only a correlation: it doesn't prove that higher shipping costs directly cause a reduction in profit margin, as other factors might influence profit. It only indicates that logistics is a related factor in profitability.



3. Evidence by Group (Higher Shipping Ratio → Worse Margin)

We divide Shipping Ratio into 5 groups (from low to high) and calculate the overall profit margin:

Shipping Ratio Group	*Total Sales*	*Total Profit*	*Profit Margin*
Lowest <0.42%	~$8.10M	~$1.83M	+22.53%
Group 2	~$3.49M	~$0.27M	+7.78%
Group 3 <1.86%	~$1.87M	~$-0.17M	-8.97%
Group 4	~$0.81M	~$-0.25M	-30.13%
Highest >3.96%	~$0.30M	~$-0.19M	-62.86%

Quantitative Conclusion:

From the "average" Shipping Ratio group onwards, profit margins start to turn negative, indicating that higher shipping costs are reducing profit efficiency.
The group with the ***highest Shipping Ratio*** has a ***strong negative profit margin*** (-62.86%), demonstrating low business efficiency for these orders.

Operational Insight:
Low-value orders with high shipping costs are ***loss-making orders***.

Common reasons may include:

Inappropriate shipping mode choice
Suboptimal packaging/container design
Prioritizing fast shipping for low-value orders

**** Recommendation***: ***Establish a Control Threshold for Shipping Ratio to Block Loss-Making Orders***
Evidence: The higher the Shipping Ratio, the more negative the profit margin (r ≈ -0.481; the highest group has a margin of -62.86%).

Action:
Set a rule: If ***Shipping Cost / Sales*** > threshold X, then:

Change the shipping mode,

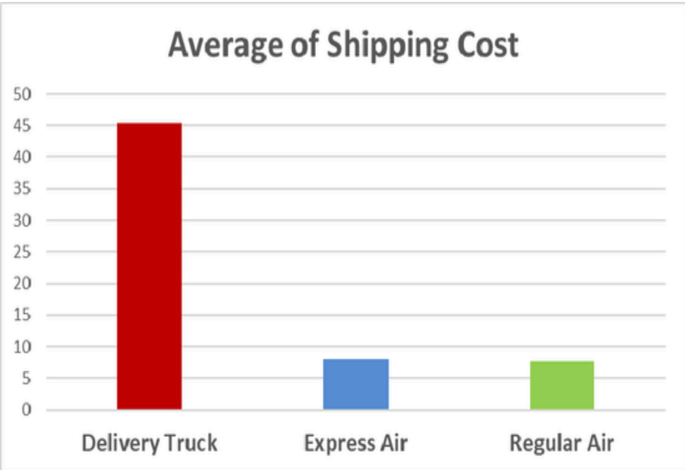
Consolidate orders,

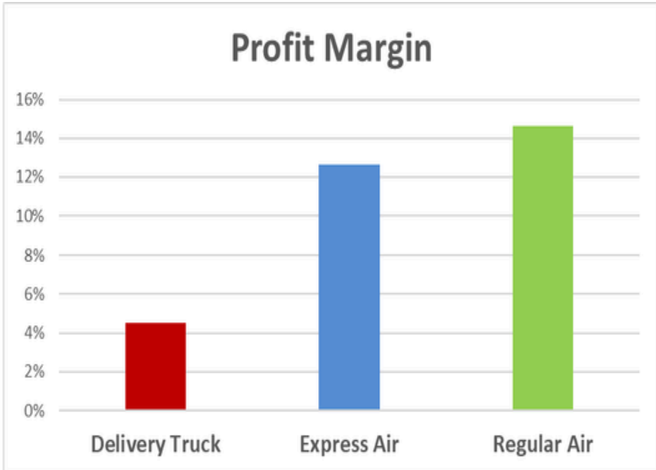
Or increase the price/reduce the discount.

4. Shipping Cost by Ship Mode: Signs of Suboptimal Optimization

We calculate the average Shipping Cost by 3 shipping methods:

Ship Mode	***Number of Orders***	***Mean Shipping Cost***	***Median***	***Profit Margin (Total)***
Regular Air	6,144	7.66	5.41	14.61%
Express Air	964	7.96	5.68	12.63%
Delivery Truck	1,118	45.44	43.75	4.52%





Observations:

Delivery Truck has an average shipping cost of ***$45.44***, which is approximately ***6 times*** higher than ***Regular Air*** and ***Express Air*** (which are around $7–8). Delivery Truck also has the highest shipping ratio at ***18%***.

Delivery Truck also has the ***lowest profit margin*** at ***4.52%***, suggesting that the choice of shipping mode directly impacts profit efficiency.

This is a sign of ***suboptimal logistics operations***, especially for orders with high shipping costs relative to their sales.

*** Recommendation**: **Restrict the Use of Delivery Truck***

Evidence: The mean shipping cost for Delivery Truck is 45.44, and the profit margin is only 4.52%.

Action:

Only allow Delivery Truck when:

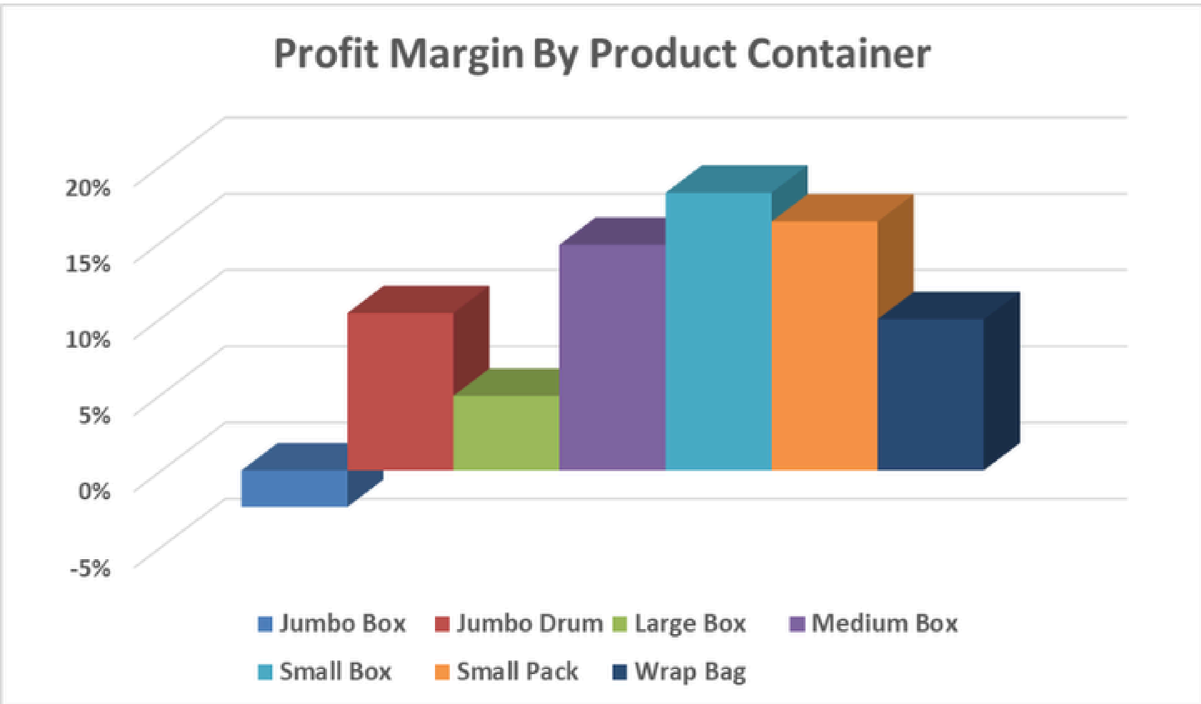
The sales value is sufficiently high (high-value orders),
The items are truly bulky and require it,
Or the expected profit margin remains positive after shipping.

5. "Hot Spots" in Operations by Product Container

Analysis by Product Container reveals:

Jumbo Box: Profit margin ≈ ***-2.43%*** (negative)

Other containers, such as ***Wrap Bags*** and ***Jumbo Boxes***, also have high Shipping/Sales ratios.



Implications:

Some types of containers are associated with low or negative profit efficiency. Possible reasons include:

Large size/volume, leading to high shipping costs

Suboptimal packaging choices

**** Recommendation***: ***Review Packaging/Containers with Low Margins (Especially Jumbo Box)***

Evidence: Jumbo Box has a negative profit margin (~ -2.43%).

Action:

Optimize the packaging method,

Switch to a different container if possible,

Or apply a shipping surcharge/minimum price for this group.

Establish Operational KPIs Linked to Profit

To ensure the focus shifts to profitability, it's crucial to develop KPIs that prioritize profit over sales.

Action:

Develop a weekly/monthly dashboard to track the following metrics:

- Percentage of orders with negative profit
- Profit Margin by Ship Mode
- Average Shipping Ratio by Region/Category
- Top 10 orders with the highest Shipping Ratios (for audit purposes)

This dashboard will help track profitability and operational efficiency, enabling the business to quickly identify and address areas that negatively impact margins. It provides clear visibility into the business's profit-focused operations and ensures that the company's strategic goals align with sustainable growth.

6. Testing Differences in Shipping Costs

Research Question: Is the average shipping cost between different Ship Modes significantly different, or is the difference due to random chance?

6.1 Hypothesis Setup

H_0 (Null Hypothesis): The mean Shipping Cost for the 3 groups (Regular Air, Express Air, Delivery Truck) are equal.
H_1 (Alternative Hypothesis): At least one group has a mean Shipping Cost that differs.

6.2 ANOVA Results (from dataset)

F-statistic ≈ 5232.07
p-value < 0.001 (very small)

Conclusion:

Reject ***H_0*** at the ***5% significance level***, meaning that the shipping costs between different Ship Modes are significantly different.
Thus, the claim "Delivery Truck is more expensive" is not just a perception, but a ***statistical fact*** supported by the data.

6.3 Operational Implications

Since ***Delivery Truck*** has the highest mean shipping cost, the company needs to establish rules for selecting the ship mode based on:

- Order value (Sales)
- Shipping Ratio
- Expected Profit Margin

V. Conclusion

This study comprehensively analyzed the sales performance, profit risks, and operational factors impacting profitability for a multi-channel retail business. Through the application of statistical methods, several key findings and actionable recommendations have emerged.

1. ***Sales Performance***: While the company achieves significant total sales, it struggles with profit volatility, largely driven by high loss-making orders. This highlights the need for a more sustainable growth model, focusing on profitability rather than just revenue generation.
2. ***Profit Analysis & Risks***: The study identified significant risks associated with discounts and shipping costs. High discount levels directly erode profit margins, while suboptimal shipping choices, particularly the use of ***Delivery Truck***, have a substantial negative impact on profitability. Furthermore, the ***Furniture*** category showed particularly low profit margins, which warrants strategic reconsideration.
3. ***Operational Factors***: The analysis of shipping costs revealed that high shipping ratios, especially for low-value orders, lead to significant losses. The study suggests optimizing shipping methods and packaging strategies to reduce costs, particularly in high-cost modes like ***Delivery Truck***. Additionally, reviewing product containers, especially ***Jumbo Boxes***, could help improve profit margins.
4. ***Data-Driven Decision Making***: Based on the findings, the study emphasizes the importance of integrating statistical analysis into daily operations. By establishing key performance indicators (KPIs) linked to profit and utilizing tools like dashboards, the company can monitor and improve its profitability over time, ensuring that growth is both sustainable and efficient.

In conclusion, while the company operates on a large scale, the findings demonstrate the importance of optimizing operational decisions, controlling costs, and shifting from a sales-driven to a profit-driven approach. Implementing these recommendations will help improve the company's profitability, reduce risks, and position it for more stable and sustainable growth in the future.